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ORIGINAL ARTICLE

Polyethylenimine-Impregnated Powdered Activated Carbon for Efficient Removal of Reactive Yellow 2 from Aqueous Solutions

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Abstract

This study optimizes polyethylenimine (PEI) impregnation on powdered activated carbon (PAC) to maximize adsorption of the anionic dye Reactive Yellow 2 (RY2). Three factors, namely PEI molecular weight (600, 800, 1200, 10,000 and 70,000 g mol^{-1}), PEI content (0.1–10 wt.%), and impregnation stirring speed (0–200 rpm), were systematically evaluated in triplicate. Under optimal conditions (PEI Mw = 800 g mol^{-1}, PEI content = 1.0 wt.%, stirring = 40 rpm), PEI-impregnated PAC achieved a maximum adsorption capacity of 477.02 mg g^{-1}, which is 7.4-fold higher than raw PAC (64.13 mg g^{-1}). FT-IR, BET, FE-SEM/EDS, and zeta potential analyses showed that although PEI impregnation reduced surface area, the introduced amine groups increased the surface charge by + 52.64 mV at pH 2, driving the enhanced adsorption. These results demonstrate that PEI-impregnated activated carbon is a strong and efficient adsorbent for RY2 removal in textile wastewater treatment.

Keywords Polyethylenimine · Activated carbon · Reactive Yellow 2 · Adsorption capacity · Wastewater treatment

Introduction

The textile dyeing industry is a major contributor to industrial water pollution, accounting for approximately 20% of global wastewater discharge and consuming nearly 10% of the world's freshwater resources [1, 2]. Among various dyes, reactive dyes such as Reactive Yellow 2 (RY2) are widely used due to their vivid color and strong bonding with textile fibers. However, their high solubility and chemical stability make them resistant to biodegradation, leading to persistent contamination in aquatic environments [3, 4]. The discharge of dye-laden wastewater can cause eutrophication, oxygen depletion, and long-term ecological risks, including bioaccumulation and toxicity [5, 6]. Even after conventional treatment, residual color often remains, necessitating additional treatment steps and increasing operational costs [7].

Various methods such as coagulation, biological degradation, and membrane separation have been employed for dye removal, but they often suffer from high energy consumption, low selectivity, and the generation of secondary pollutants such as sludge [8–10]. Adsorption has gained attention as an attractive alternative due to its simplicity, effectiveness, and minimal by-product formation [11]. Among adsorbents, powdered activated carbon (PAC) is widely used owing to its large surface area and broad adsorption range. However, raw PAC shows limited performance for anionic dyes due to its negatively charged surface under neutral to basic pH and the lack of electrostatically interactive functional groups [12]. Its fine particle size also hampers recovery and reuse, limiting its practical applicability in large-scale processes [13].

To address these limitations, surface modification of PAC with functional polymers has been widely investigated. Among numerous polymers, polyethylenimine (PEI), a highly branched polyamine, introduces amine-rich functional groups that become protonated under acidic conditions, generating positively charged sites for electrostatic interactions with anionic dyes [14, 15]. While various studies have demonstrated the potential of PEI-modified adsorbents, many have used biomass-derived or non-powdered substrates, reported moderate adsorption capacities

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(< 300 mg g^{-1}), or required complex multi-step modifications that hinder scalability [16–18]. In addition, essential parameters such as PEI molecular weight, loading concentration, and mixing conditions during impregnation have rarely been systematically optimized.

In this study, we report a simple, one-step impregnation method for fabricating PEI-impregnated PAC (PEI/PAC) using SPS-100, a commercially available powdered activated carbon. The modification was designed to enhance the electrostatic affinity of PAC toward anionic dyes by introducing amine functional groups. Key variables, including PEI molecular weight, PEI content, and stirring speed, were systematically optimized to obtain an effective adsorbent. The resulting material was evaluated in terms of dye adsorption performance, desorption efficiency, and reusability under batch-mode conditions.

Overall, this work presents a practical and tunable surface modification strategy for PAC and aims to provide insights into the rational design of amine-functionalized adsorbents for enhanced dye removal in wastewater treatment systems.

Experimental

Materials

SPS-100 was supplied by Samchully Activated Carbon Co., Ltd. (Geumsan, Korea). Branched PEI with a molecular weight (M_w) of 800 (98%) and RY2 (dye content ≥ 60%) were purchased by Sigma-Aldrich Korea Ltd. (Seoul, Korea). Branched PEIs of different molecular weights (M_w = 600, 1200, 10,000 and 70,000, 98–100%) were provided by Nippon Shokubai Co. Ltd. (Osaka, Japan). HCl and NaOH were offered by Daejung Chemicals & Metals Co., Ltd. (Siheung, Korea).

Fabrication of PEI-Impregnated PAC

Various PEI/PACs were prepared by a one-step impregnation of 2 g PAC in 50 mL aqueous PEI solution. All experiments were performed in triplicate ($n = 3$), and results are reported as mean ± standard error. Impregnation parameters were selected based on preliminary experiments to optimize polymer–carbon interactions.

Three variables were investigated. First, the effect of PEI molecular weight was assessed by reacting PAC with 5 wt.% PEI solutions of different chain lengths (600, 800, 1200, 10,000 and 70,000 g mol^{-1}) at 160 rpm and 25 °C for 72 h to determine how polymer chain length influences pore accessibility. Second, the impact of PEI content was evaluated by treating PAC with PEI (M_w = 800 g mol^{-1}) solutions at 0.1, 1, 5, and 10 wt.% under the same stirring and temperature conditions for 72 h to establish the optimal polymer loading. Third, mass-transfer effects were examined by impregnating PAC in a 1 wt.% PEI solution (M_w = 800 g mol^{-1}) at 25 °C for 72 h under stirring speeds of 0, 40, 80, 120, 160, and 200 rpm.

After impregnation, each suspension was centrifuged at 2000 rpm for 10 min, washed three times with distilled water, and filtered. The wet solids were freeze-dried at −50 °C under 0.1 mbar for 24 h. Optimal conditions (PEI M_w = 800 g mol^{-1}, PEI content = 1 wt.%, stirring = 40 rpm) were selected based on the highest adsorption capacity in the evaluation of each influencing factor, and the sample prepared under this optimal condition was named 1% PEI/PAC-800 and was used for all subsequent adsorption experiments.

Characterizations

A Fourier transform infrared spectrometer (FT-IR, FT/IR-8X, Jasco, Japan) was used to record the FT-IR spectra of PAC and PEI/PAC in the range of 4000–500 cm^{-1}. A 3Flex surface characterization analyzer (Micromeritics, USA) was used to collect the nitrogen adsorption–desorption isotherms of raw PAC and 1% PEI/PAC-800. A field emission scanning electron microscope with energy dispersive spectroscopy (FE-SEM/EDS, JSM-7610F, Jeol, Japan) was used to record the surface morphologies of activated carbons. A zeta potential analyzer (ELSZ-2000, Otsuka, Japan) was used to analyze the zeta potential values of raw PAC and 1% PEI/PAC-100 at different pH values.

Adsorption Experiments

The stock solution of RY2 was prepared according to our previous method [19]. In all batch adsorption experiments, 30 mg of adsorbent and 30 mL of RY2 solution were mixed in a 50-mL conic tube and incubated in a shaker at 160 rpm. Unless otherwise noted, the pH, shaking speed, and temperature for all adsorption experiments were 2, 160 rpm, and 25 °C.

pH edge experiments were separately performed over the pH range of 2–12 (adjusted using 0.1 M HCl or NaOH) with an initial RY2 concentration of 485.59 mg L^{-1}. Samples were equilibrated for 7 d to assess pH-dependent adsorption behavior. Kinetic experiments were conducted at an initial RY2 concentration of 504.25 mg L^{-1}. Aliquots were withdrawn at predetermined time intervals until no further change in concentration was observed, indicating equilibrium. The residual RY2 concentration was measured at 404 nm using a UV–Vis spectrophotometer (X-ma 3000 pc, Human, Korea). The kinetic data were analyzed using pseudo-first-order and pseudo-second-order models to determine the adsorption rate parameters. Isotherm experiments were conducted by varying the initial RY2 concentrations from 30 to 1000 mg L^{-1} under identical conditions until

equilibrium (typically 7 d) was reached. The equilibrium data were fitted to the Langmuir and Freundlich models to evaluate the maximum adsorption capacity and surface heterogeneity.

The dye uptake, q (mg g^{-1}), was calculated using the following equation:

$$q = \frac{C_i V_i - C_f V_f}{m} \quad (1)$$

where C_i, C_f (mg L^{-1}) and V_i, V_f (L) are the RY2 concentration and the solution volume before and after adsorption, and m (g) is the weight of the adsorbent.

Reusability Studies

First, 30 mg of adsorbent and 30 mL of RY2 (100 mg L^{-1}) were mixed at 25 °C for 4 h to obtain RY2-loaded adsorbent. After washing three times with distilled water, the RY2-loaded adsorbent was eluted with 30 mL of NaOH solution (0.01 mol L^{-1}) for 2 h, and the adsorption–desorption was repeated three times to confirm the recyclability in removing RY2. The RY2 concentration was measured by UV–Vis spectrophotometer, and the desorption rate was calculated by Eq. (2).

$$\text{Desorption rate (\%)} = \frac{\text{Desorbed RY2 in each cycle (mg)}}{\text{Adsorbed RY2 in each cycle (mg)}} \times 100 \quad (2)$$

Results and Discussion

Optimization of PEI Impregnation for Maximum RY2 Adsorption

To determine the optimal modification parameters for RY2 adsorption, raw PAC and various PEI/PACs were evaluated for three variables: PEI molecular weight, PEI content, and stirring speed during impregnation.

Influence of PEI Molecular Weight

The adsorption capacities of raw PAC and PAC modified with PEI of varying molecular weights (600, 800, 1200, 10,000 and 70,000 g mol^{-1}) are compared in Fig. 1a. Modification enhanced RY2 uptake in all cases; the trend in capacity was: PEI/PAC-800 > PEI/PAC-600 > PEI/PAC-1200 > PEI/PAC-10000 ≈ PEI/PAC-70000 > raw PAC. The number following PEI/PAC indicates the PEI molecular weight. PEI of 800 g mol^{-1} provides an optimal balance between chain length (enabling deep pore penetration) and amine-group density (maximizing electrostatic binding). Higher PEI molecular weights (≥ 1200 g mol^{-1}) are likely to be sterically hindered, which can limit pore access and reduce the surface area with amine groups available for binding to RY2 [20]. In contrast, PEI-600 is unhindered but has fewer functional groups than PEI-800, resulting in a slightly reduced adsorption capacity. These results indicate an optimal PEI molecular weight around 800 g mol^{-1} for maximizing dye uptake.

Influence of PEI Content

Figure 1b shows the effect of PEI content on the adsorption capacity of RY2, and the corresponding change in nitrogen content was also analyzed by EDS (Fig. S1). As the PEI loading increased from 0.1 to 1.0 wt.%, the RY2 uptake increased rapidly, reaching a maximum at 1.0 wt.%. This is attributed to the electrostatic interaction between the amine groups introduced by PEI and the anionic RY2 [21, 22], as confirmed by the nitrogen content trend. However, when the PEI content exceeds 1.0 wt.%, the adsorption capacity tends to decrease, likely due to excessive PEI blocking micropores and mesopores, thereby reducing the effective surface area for dye adsorption [23]. This finding is consistent with the

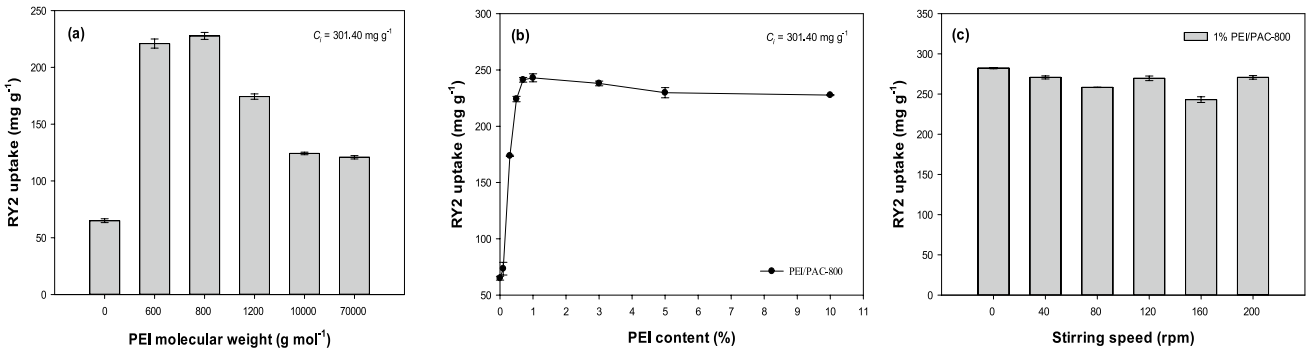


Fig. 1 Effects of **a** PEI molecular weight, **b** PEI content, and **c** stirring speed on the adsorption of RY2. The experimental conditions were: PEI/PAC-800, C_i = 301.4 mg L^{-1}, pH = 2, and contact time = 7 days

study by Hamdy et al. [24], which reported that an excess of high-molecular weight PEI can obstruct pore structures and active sites, resulting in reduced adsorption efficiency. Although the nitrogen content remains nearly constant beyond 1.0 wt.%, the adsorption capacity declines. This highlights the need for an optimal PEI content to prevent pore blockage and polymer aggregation [25]. Therefore, 1.0 wt.% PEI provides the optimal trade-off between functional group availability and accessible surface area for RY2 adsorption.

Influence of Stirring Speed

The effect of stirring speed during the impregnation of 1% PEI/PAC-800 on RY2 adsorption capacity is shown in Fig. 1c. The highest adsorption capacity was observed at 0 rpm; however, the overall impact of stirring speed on performance was limited. Static impregnation without mixing may lead to issues with reproducibility and non-uniform polymer distribution. In contrast, stirring at 40 rpm provided similar adsorption capacity while ensuring uniform dispersion, making it a more practical and reliable condition. Therefore, 40 rpm was selected as the optimal condition, considering adsorption performance, reproducibility, and experimental stability.

Summary of Optimal Modification Conditions

Under the conditions studied, the highest RY2 adsorption capacity was achieved with a PEI molecular weight of 800 g mol^{-1}, a PEI content of 1.0 wt.%, and a stirring speed of 40 rpm. This optimal performance results from a well-balanced combination of sufficient functional group density (i.e., available amine groups), effective pore accessibility with minimal steric hindrance, and adequate mass-transfer efficiency due to proper mixing. The decline in adsorption observed at higher PEI molecular weights and contents highlights the importance of avoiding excessive steric hindrance and pore blockage during the design of polymer-modified adsorbents.

Characterization

The BET analysis results (Fig. 2a, b) show that although the specific surface area of PAC decreased significantly after PEI impregnation, the adsorption capacity of 1% PEI/PAC-800 improved remarkably (Fig. 1a). While activated carbon's surface area typically governs its adsorption performance by providing accessible active sites, this outcome highlights that surface chemistry can also play a dominant role. A similar phenomenon has been reported by Bai et al. [25], who observed enhanced CO_2 capture using PEI-impregnated resins despite surface area reduction.

The pore size distribution (Fig. 2c, d) helps explain this behavior. For 1% PEI/PAC-800, pores in the 4–8 nm range nearly disappeared, likely due to blockage by the impregnated PEI. This is consistent with the molecular length of PEI ($M_w = 800$ g mol^{-1}), whose longest segment is approximately 2.7 nm (Scheme 1). While smaller PEI chains may partially penetrate mesopores, their branched structure can cause pore constriction rather than complete occlusion, thereby altering the porous architecture. Nevertheless, the introduced amine groups serve as new active sites for dye uptake, enhancing adsorption through electrostatic attraction in addition to physical adsorption. Hence, the net effect of PEI loading is a trade-off: reduced surface area but increased functional group density.

In summary, the decreased surface area caused by partial pore blockage is offset by the introduction of PEI-derived amine groups, which provide additional adsorption sites. The balance between structural modification and chemical functionality contributes to the superior performance of 1% PEI/PAC-800. These findings are in agreement with the work of Xu et al. [26] on PEI-modified mesoporous materials, where dual physical and chemical adsorption mechanisms were found to synergistically enhance performance.

FT-IR spectroscopy (Fig. 3) provides further evidence of successful PEI impregnation. The raw PAC shows a broad O–H stretching band at 3000–3500 cm^{-1}, corresponding to surface hydroxyl groups, along with peaks at 1613 cm^{-1} (C = O stretching) and 1166 cm^{-1} (C–O stretching) due to oxidation during activation [27]. After PEI modification, the broad band in the 3000–3500 cm^{-1} region shifts and overlaps with N–H stretching vibrations, indicating the presence of amine groups. In addition, a distinct peak at 1541 cm^{-1} corresponds to N–H bending, further confirming PEI incorporation [28]. The reduced intensity of the O–H band suggests that some surface hydroxyls may have reacted with or been blocked by PEI.

SEM and EDS analyses were conducted to examine changes in surface morphology and elemental composition (Fig. S2). Both raw PAC and 1% PEI/PAC-800 exhibit irregular, porous structures characteristic of wood-based activated carbon [29]. Following PEI impregnation, the surface became slightly smoother, though the overall morphology remained similar. EDS results showed that raw PAC consisted primarily of carbon with trace oxygen, while 1% PEI/PAC-800 exhibited clear nitrogen signals, attributed to amine groups from PEI (Figs. S3, S4), confirming successful chemical modification.

After RY2 adsorption, the surface of the loaded 1% PEI/PAC-800 appeared rougher (Fig. 4), suggesting dye molecule attachment. EDS spectra of the RY2-loaded sample revealed the presence of sulfur and chlorine, alongside carbon, oxygen, and nitrogen. These elements originate from the RY2 dye (Fig. S5 for its chemical structure), validating

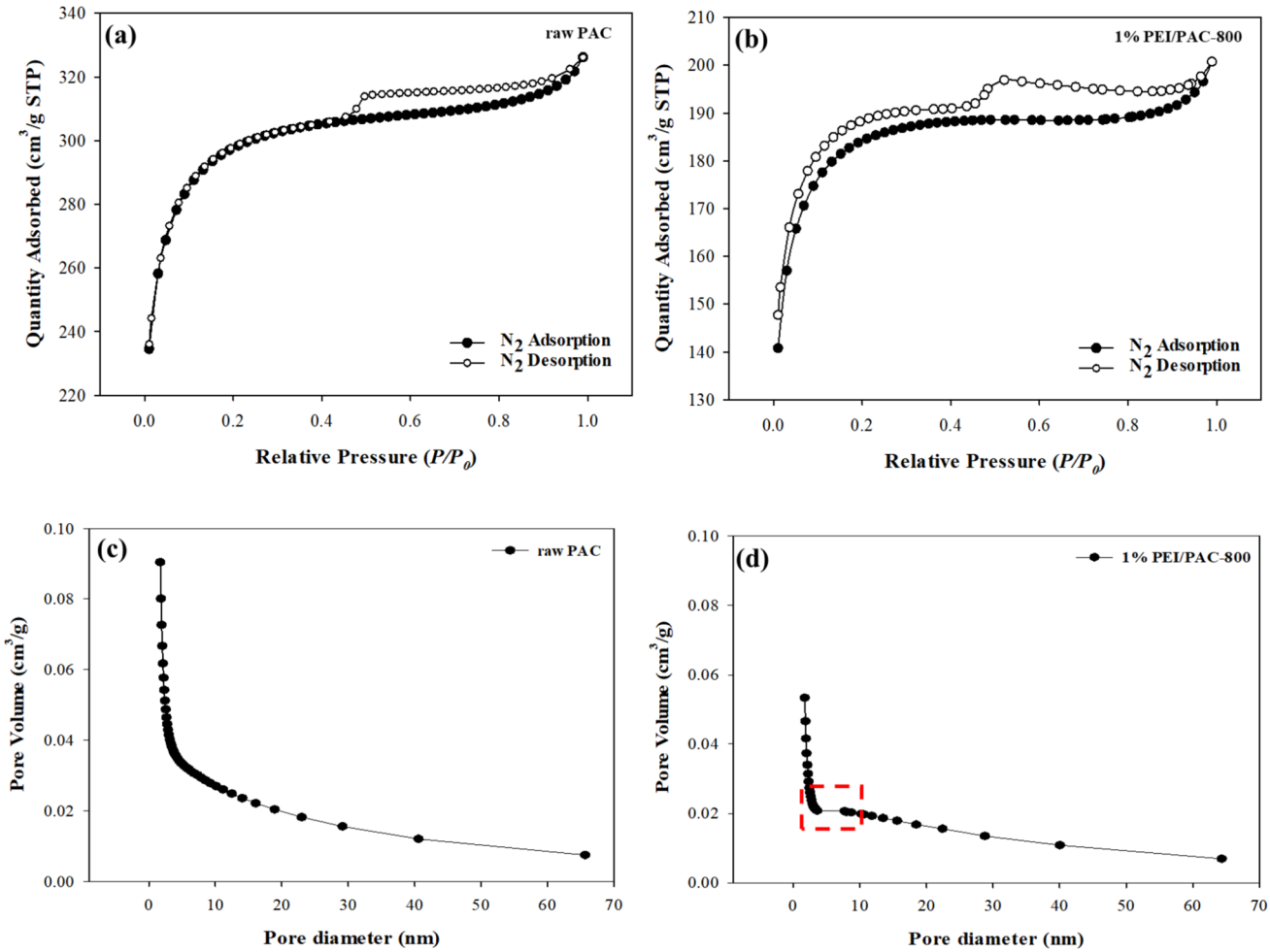


Fig. 2 **a**, **b** Nitrogen adsorption–desorption isotherms and **c**, **d** pore size distributions of raw PAC and 1% PEI/PAC-800, respectively

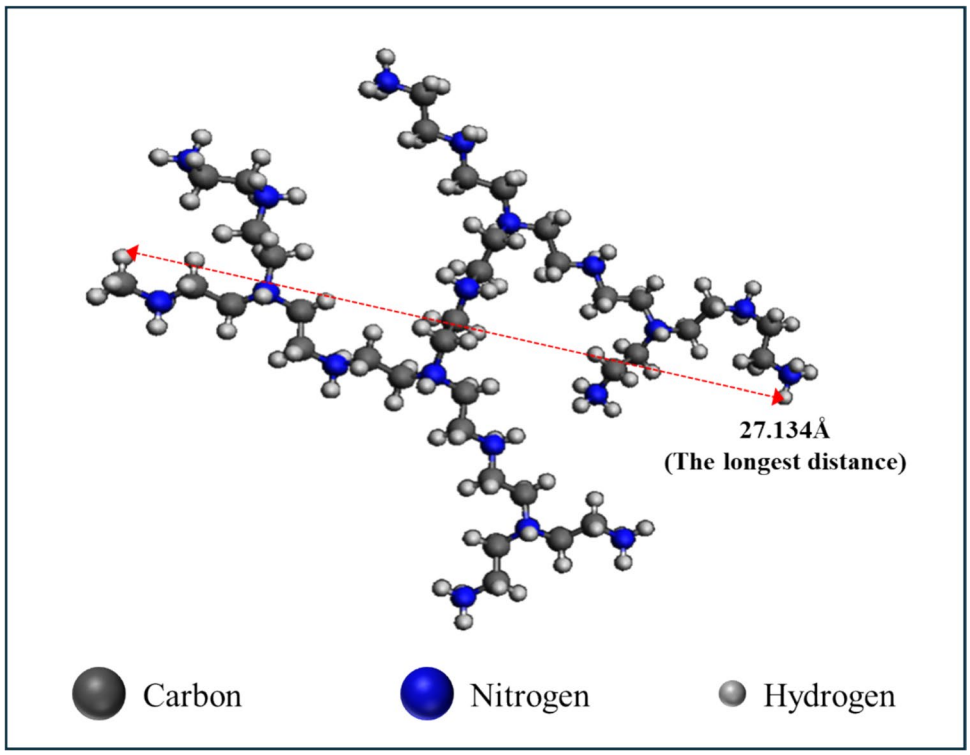
its successful adsorption onto the PEI-impregnated PAC. Collectively, these results demonstrate that PEI impregnation introduces functional groups that not only modify surface chemistry but also significantly enhance dye adsorption performance.

Effect of pH

Figure 5a depicts the effect of pH on the adsorption of RY2 by raw PAC and 1% PEI/PAC-800. Compared to PAC, 1% PEI/PAC-800 showed a significant change in the adsorption of RY2 with pH. At pH 2, 1% PEI/PAC-800 has an adsorption capacity of 448.2 mg/g, approximately 5.6 times higher than that of raw PAC (79.5 mg/g). However, the adsorption capacity drops sharply to 130.1 mg/g at pH 4 and further decreases to 56.2 mg/g above pH 7. In contrast, raw PAC maintains a relatively stable adsorption capacity in the range of 60–80 mg/g across all pH levels, albeit at lower performance.

This strong pH sensitivity of 1% PEI/PAC-800 is closely related to its surface charge characteristics, as illustrated in the zeta potential analysis (Fig. 5b). At pH 2, its surface exhibits a high positive zeta potential (+52.6 mV) due to the protonation of amine groups ($–NH_3^+$), which facilitates strong electrostatic attraction with anionic dye molecules. As the pH increases, the surface charge decreases steadily, approaching neutrality at around pH 10 and becoming negative at pH 12 (–36.9 mV), indicating a reversal in surface charge. The point of zero charge (pH_{pzc}) is determined to be 10.1, which is similar to previous studies on PEI-based materials (~10.8) [30]. The diminished surface charge at higher pH levels weakens electrostatic interactions, resulting in lower dye adsorption efficiency.

Raw PAC exhibits a pH-dependent surface charge behavior, being slightly positive under acidic conditions and increasingly negative with rising pH. However, its surface potential is significantly lower than that of 1% PEI/PAC-800, reflecting much weaker electrostatic interaction with anionic dye molecules. Consequently, RY2



Scheme 1 Molecular structure of branched polyethylenimine ($M_w = 800$), highlighting the longest linear segment

adsorption by raw PAC is primarily driven by physical mechanisms such as π–π stacking, hydrophobic interactions, and micropore diffusion, which account for its relatively consistent adsorption capacity across the pH range [31].

In summary, 1% PEI/PAC-800 exhibits enhanced dye adsorption under acidic conditions due to strong electrostatic attraction from its highly positive surface charge. However, its performance diminishes at higher pH due to progressive deprotonation of surface amines. These results suggest that PEI-impregnated PAC is particularly effective for dye removal in acidic wastewater, although further surface modifications may be necessary to maintain performance under neutral or alkaline conditions.

Adsorption Isotherms and Modeling

Adsorption isotherms were conducted to evaluate the adsorption capacity and mechanism of raw PAC and 1% PEI/PAC-800 for RY2 removal. The experimental data were analyzed by fitting to both the Langmuir and Freundlich models, which represent different adsorption mechanisms based on surface homogeneity (Table 1 and Fig. 6a). The Langmuir model assumes monolayer adsorption on a homogeneous surface with identical binding sites and no interactions between adsorbed molecules [32]. The Langmuir equation is represented as:

$$q_e = \frac{q_{max} b_L C_e}{1 + b_L C_e} \tag{3}$$

where q_e (mg g^{-1}) is the amount of adsorbate adsorbed per unit weight of adsorbent, C_e (mg L^{-1}) is the equilibrium concentration of the adsorbate, q_{max} (mg g^{-1}) is the maximum adsorption capacity, and b_L (L mg^{-1}) is the Langmuir constant related to the affinity of binding sites. In contrast, the Freundlich model describes multilayer adsorption on heterogeneous surfaces with varying adsorption energies. It is more applicable when the surface exhibits a non-uniform distribution of adsorption heat and site affinity. The Freundlich equation is expressed as:

$$q_e = K_F C_e^{1/n} \tag{4}$$

where K_F ((mg g^{-1}) (L mg^{-1})$^{1/n}$) is the Freundlich constant indicative of adsorption capacity, and $1/n$ is a measure of adsorption intensity or surface heterogeneity.

To further elucidate the adsorption behavior, the experimental data were additionally fitted to the Sips and Redlich–Peterson (R–P) models (Table 1 and Fig. 6b). The Sips model, also referred to as the Langmuir–Freundlich model, combines the features of both Langmuir and Freundlich equations, effectively describing heterogeneous systems that approach Langmuir behavior at high adsorbate concentrations. The Sips equation is given as:

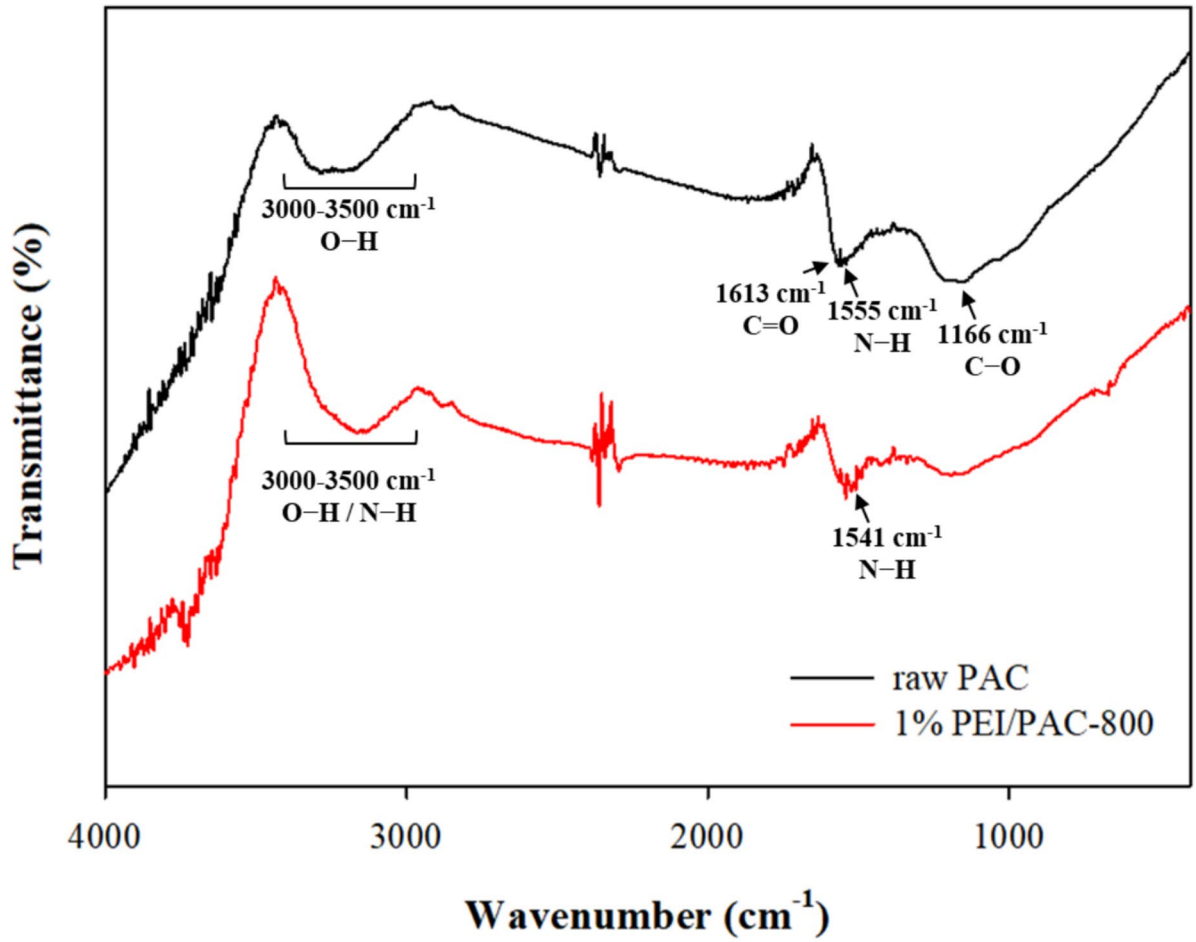


Fig. 3 FT-IR spectra of raw PAC and 1% PEI/PAC-800, confirming the presence of functional groups introduced by PEI impregnation

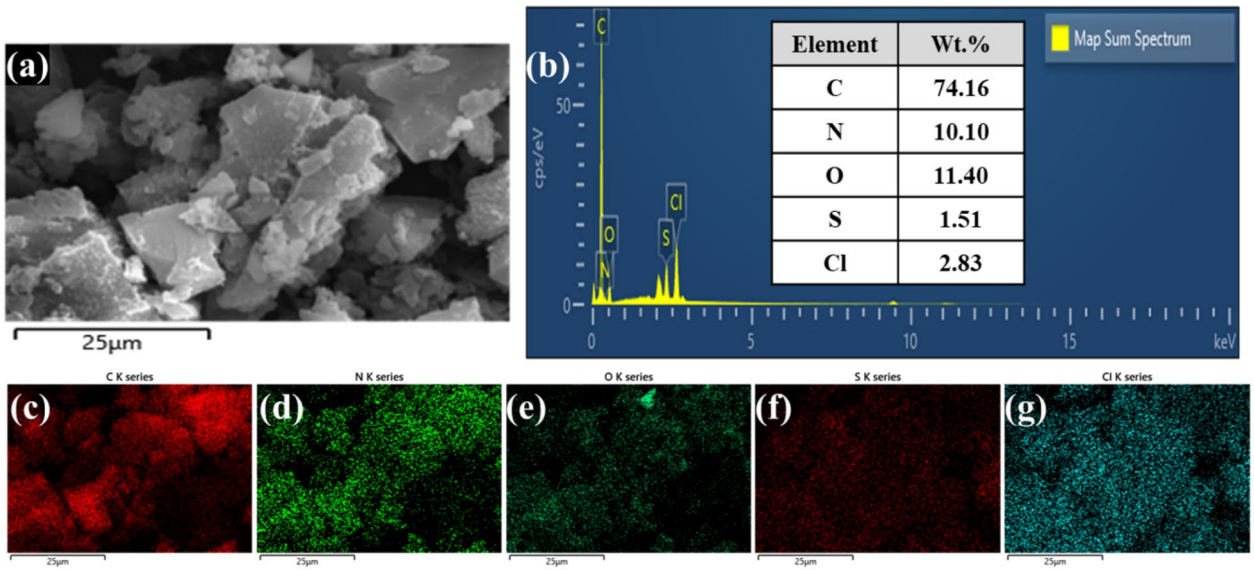


Fig. 4 SEM image and EDS analysis of RY2-loaded 1% PEI/PAC-800: **a** surface morphology, **b** EDS spectrum and elemental composition; **c**–**g** elemental mapping of C, N, O, S, and Cl

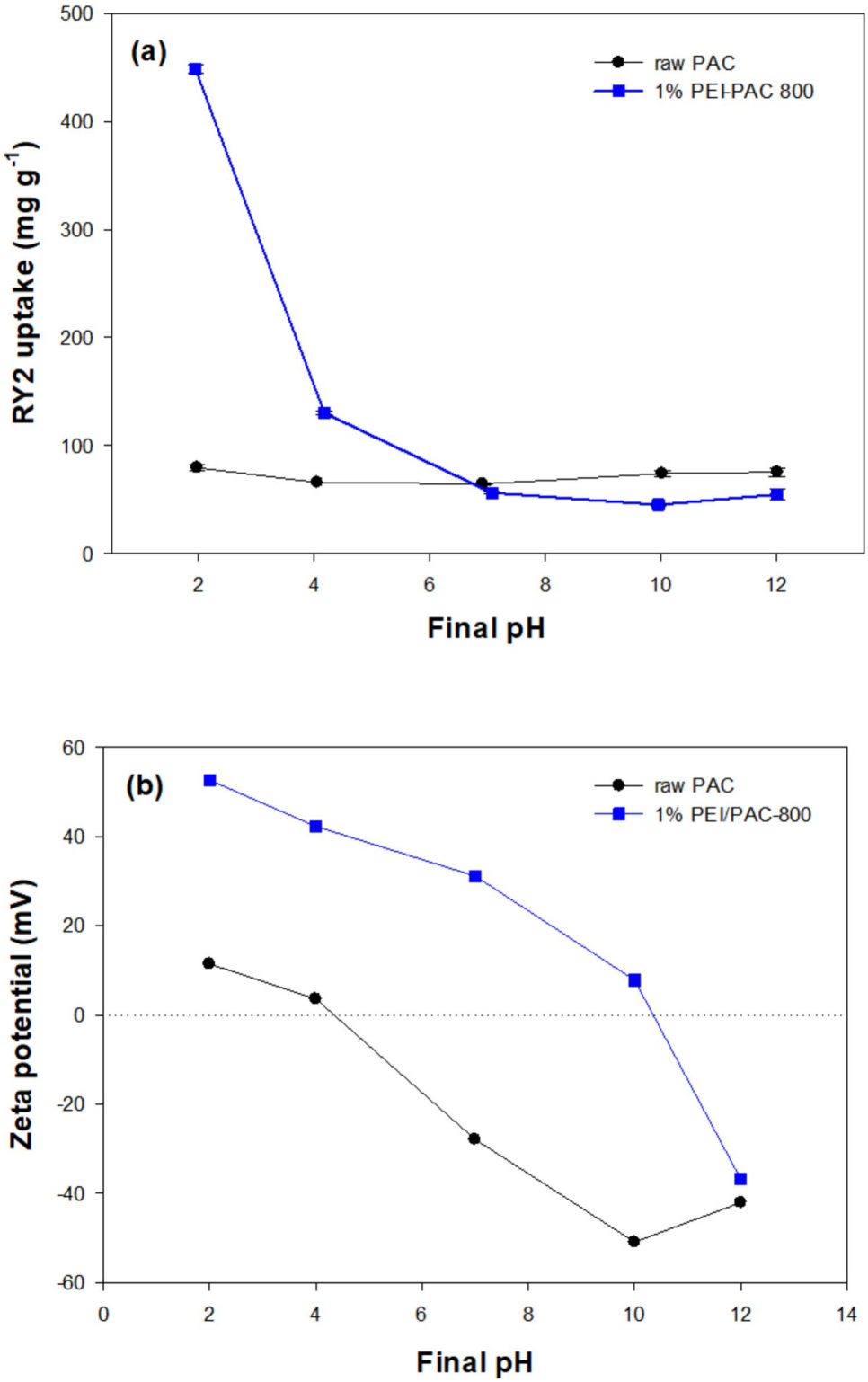


Fig. 5 Effect of pH on **a** RY2 adsorption capacity and **b** zeta potential of raw PAC and 1% PEI/PAC-800 (C_i = 485.59 mg L^{-1}, contact time = 7 days)

$$q_e = \frac{q_{max}(K_S C_e)^{n_S}}{1 + (K_S C_e)^{n_S}} \quad (5)$$

where q_{max} (mg g^{-1}) is the maximum adsorption capacity, K_S (L mg^{-1}) is the Sips equilibrium constant, and n_S is the heterogeneity factor; when $n_S = 1$, the model reduces to the Langmuir form. The R–P model is an empirical isotherm that combines elements of both Langmuir and Freundlich models and can describe adsorption over a wide concentration range. It is expressed as:

Table 1 Isotherm model parameters for RY2 adsorption onto activated carbons

Isotherm model	Parameter	Raw PAC	1% PEI/PAC-800
Langmuir	q_{max} (mg g^{-1})	64.13	477.02
	b_L (L mg^{-1})	0.781	0.692
	R^2	0.982	0.906
Freundlich	K_F (L g^{-1})	28.66	186.98
	n_F	6.652	5.870
	R^2	0.906	0.662
Sips	q_{max} (mg g^{-1})	64.14	438.18
	K_S (L mg^{-1})	0.782	2.294
	n_S	0.998	3.577
	R^2	0.973	0.973
Redlich–Peterson	K_R (L g^{-1})	70.23	313.96
	α_R (L mg^{-1})	1.343	0.605
	β	0.961	1.019
	R^2	0.981	0.902

$$q_e = \frac{K_R C_e}{1 + \alpha_R C_e^{\beta}} \tag{6}$$

where K_R (L g^{-1}) and α_R (L mg^{-1}) are R–P constants, and β ($0 \le \beta \le 1$) is the heterogeneity exponent. When $\beta = 1$, the equation simplifies to the Langmuir equation, while value of β approaching 0 reflects Freundlich-like behavior.

For raw PAC, the Langmuir model provided an excellent fit ($R^2 = 0.982$), yielding a maximum adsorption capacity (q_{max}) of 64.13 mg g^{-1} and a Langmuir constant (b_L) of 0.781. The Freundlich model showed moderate agreement ($R^2 = 0.906$), with $K_F = 28.66$ and $1/n = 0.1503$, indicating relatively low adsorption intensity. In the case of 1% PEI/PAC-800, the Langmuir model demonstrated a superior fit ($R^2 = 0.906$), with a significantly higher $q_{max} = =477.02$ mg g^{-1} and $b_L = 0.692$, reflecting favorable and reversible binding. Although the Freundlich model showed a higher $K_F = 186.98$, it was not suitable for fitting the 1% PEI/PAC-800 due to the low R^2 value (0.662). These results confirm that monolayer adsorption dominates both adsorbents and that the Langmuir model more accurately describes their behavior, particularly for the PEI-impregnated PAC. The enhanced adsorption performance of 1% PEI/PAC-800 can be attributed to introducing amine groups through PEI impregnation, which enables strong electrostatic and chemical interactions with anionic dye molecules. The notably higher q_{max} and K_F values of 1% PEI/PAC-800 reaffirm that PEI impregnation substantially enhances both the adsorption capacity and binding affinity by increasing the number of active sites and promoting stronger interactions with RY2.

Additional fitting using the three-parameter Sips and R–P models provided additional insights into surface heterogeneity and deviations from ideal monolayer adsorption. For raw PAC, both models exhibited high correlation ($R^2 = 0.973$ for Sips and 0.981 for R–P), with heterogeneity parameters ($n_S = 0.998$, $\beta = 0.961$) confirming a relatively uniform surface consistent with the Langmuir-type behavior. In contrast, 1% PEI/PAC-800 showed markedly higher heterogeneity ($n_S = 3.577$, $\beta = 1.019$), and the Sips model yielded the best fit ($R^2 = 0.973$). This indicates that PEI impregnation introduces a diverse range of binding sites and enables both electrostatic and chemical interactions. Overall, while monolayer adsorption remains the main mechanism, the enhanced performance of the PEI-impregnated adsorbent is also influenced by increased surface heterogeneity and dual adsorption mechanisms.

Notably, as shown in Table 2, the maximum adsorption capacity of 1% PEI/PAC-800 (477.01 mg g^{-1}) outperforms many other reported adsorbents, including commercial activated carbon (Calgon, 209.5 mg g^{-1}) and a variety of biomass-based adsorbents [28, 33–38]. While PEI/APTES-MWCNT demonstrates an even higher adsorption capacity (742.4 mg g^{-1}), the synthesis process is much more complex, requiring several steps [39]. In contrast, the simple PEI impregnation process employed here offers an efficient, cost-effective route to high-performance adsorbents suitable for large-scale wastewater remediation.

Adsorption Kinetics and Modeling

The adsorption kinetics of RY2 onto raw PAC and 1% PEI/PAC-800 were investigated to evaluate adsorption mechanisms and rate behaviors using pseudo-first-order and pseudo-second-order models (Table 3 and Fig. 6b). The kinetic equations are expressed as follows:

$$\text{Pseudo-first-order model: } \ln(q_e - q_t) = lnq_e - k_1 t \tag{7}$$

$$\text{Pseudo-second-order model: } \frac{t}{q_t} = \frac{1}{(k_2 q_e^2)} + \frac{t}{q_e} \tag{8}$$

where q_e and q_t (mg g^{-1}) are the amount of dye adsorbed at equilibrium and time t, respectively, k_1 (min^{-1}) is the pseudo-first-order rate constant, and k_2 (g mg^{-1} min^{-1}) is the pseudo-second-order rate constant.

Fitting to the pseudo-first-order model yielded an R^2 value of 0.704 for raw PAC with a theoretical capacity (q_e) of 68.62 mg/g and $k_1 = 1.857$ min^{-1}, while 1% PEI/PAC-800 showed a slightly higher R^2 of 0.767 with $q_e = 376.18$ mg g^{-1} and $k_1 = 0.105$ min^{-1}. Although 1% PEI/PAC-800 exhibited significantly higher capacity, the lower rate constant suggests slower initial uptake. In contrast, the pseudo-second-order model provided better correlation, with R^2 values of 0.774 and 0.849 for raw PAC and 1% PEI/PAC-800, respectively. The estimated equilibrium capacities were 70.72 and

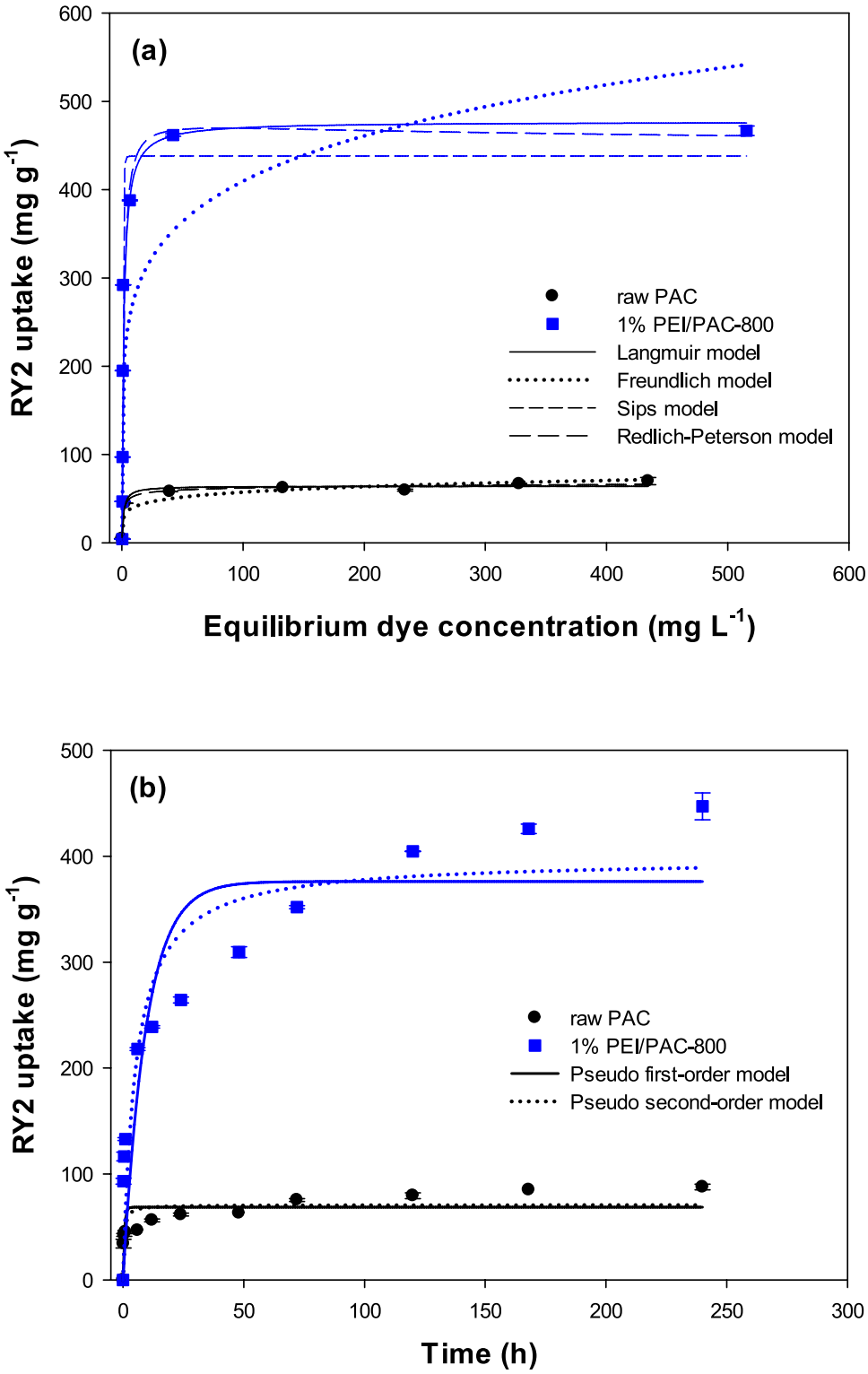


Fig. 6 **a** Adsorption isotherms of RY2 on raw PAC and 1% PEI/PAC-800 fitted to the Langmuir, Freundlich, Sips, and Redlich–Peterson models ($C_i = 30 - 1000$ mg L^{-1}, pH = 2, contact time = 7 days), **b** adsorption kinetics fitted to pseudo-first-order and pseudo-second-order models ($C_i = 504.25$ mg L^{-1}, pH = 2)

397.39 mg g^{-1}, while the corresponding k_2 values were 0.0374 and 0.0005 g mg^{-1} min^{-1}. These results indicate that the adsorption kinetics are better described by the pseudo-second-order model. Although pseudo-second-order kinetics are frequently linked to chemisorption, the dominant mechanism in this study is better attributed to site-specific electrostatic interactions between the anionic dye and the protonated amine groups introduced by PEI.

The initial adsorption rate (h) is further evaluated using the following relation:

$$h = k_2 q_e^2 \quad (9)$$

The calculated values of h were 187.02 mg g^{-1} min^{-1} for raw PAC and 78.96 mg g^{-1} min^{-1} for 1% PEI/PAC-800. Although raw PAC exhibited a higher initial uptake rate, its overall adsorption capacity remained low. In contrast, the PEI-modified adsorbent displayed a slower rate but substantially higher capacity, underscoring the trade-off between kinetic accessibility and binding affinity.

A detailed comparison of kinetic behavior between raw PAC and 1% PEI/PAC-800 illustrates the interplay between pore structure and surface functionality. As shown in Table 4, raw PAC, characterized by an extremely high specific surface area (1112.7 m^2 g^{-1}) and dominant microporous features (Horvath–Kawazoe median pore width = 0.77 nm), exhibited a rapid initial adsorption rate but limited overall uptake. This limitation can be attributed to the mismatch between the narrow micropores and the bulky molecular size of RY2 (Mw = 831.02 g mol^{-1}), which impedes internal diffusion and restricts adsorption to external surfaces or a minor fraction of accessible mesopores. Nevertheless, the presence of mesopores, though limited, may still facilitate partial intra-particle diffusion, providing supplementary pathways for RY2 migration into the adsorbent surface.

Notably, the external surface area of raw PAC was negligible, indicating that most of the surface was confined within micropores inaccessible to large dye molecules. On the other hand, 1% PEI/PAC-800 exhibited a markedly higher adsorption capacity despite a reduced adsorption rate. The enhancement is primarily ascribed to the incorporation of abundant protonated amine groups that enable strong electrostatic interactions with the anionic dye. Moreover, the external surface area of 1% PEI/PAC-800 (66.4 m^2 g^{-1}) was significantly greater than that of raw PAC, facilitating effective surface binding even in the absence of deep pore penetration. However, the impregnation of PEI also partially

Table 2 Comparison of maximum adsorption capacities (q_{max}) for RY2 on various adsorbent materials

Adsorbent	q_{max} (mg g^{-1})	References
Polyethylenimine modified calcium silicate hydrate	235.0	Kang et al., [28]
Commercial activated carbon (Calgon)	209.5	Al-Degs et al., [33]
Activated sludge	333.3	Aksu, [34]
Corynebacterium glutamicum biomass	178.5	Won and Yun, [35]
Polysulfone/bacterial biomass composite fiber	153.2	Park et al., [36]
Magnetic carbon composite	112.6	Mello et al., [37]
Immobilized *Gibberella fujikuroi* on maize tassel biomatrix	90.0	Celik et al., [38]
Polyethylenimine-3-aminopropyltriethoxysilane-MWCNT	742.4	Wang and Won, [39]
Raw PAC	64.13	This study
1% PEI/PAC-800	477.02	This study

Table 3 Kinetic model parameters for RY2 adsorption onto activated carbons

Adsorbent	Pseudo-first-order			q_2 (mg g^{-1})	Pseudo-second-order		
	q_1 (mg g^{-1})	k_1 (L min^{-1})	R^2		k_2 (g mg^{-1} min^{-1})	h (mg g^{-1} min^{-1})	R^2
Raw PAC	68.62	1.857	0.704	70.72	0.0374	187.02	0.774
1% PEI/PAC-800	376.18	0.105	0.767	397.39	0.0005	78.96	0.849

Table 4 Comparison of BET surface area and pore structure parameters for raw PAC and 1% PEI/PAC-800

Property	Raw PAC	1% PEI/PAC-800
BET surface area (m^2 g^{-1})	1112.8	690.1
Langmuir surface area (m^2 g^{-1})	1370.6	830.2
Micropore area (m^2 g^{-1})	1261.5	623.7
Micropore volume (cm^3 g^{-1})	0.4144	0.2569
External surface area (m^2 g^{-1})	–	66.4
BJH pore volume (1.7–300 nm, cm^3 g^{-1})	0.0802	0.0534
BJH average pore diameter (nm)	3.27	3.26
H–K median pore width (nm)	0.77	0.7722
Adsorption average pore diameter (BET, nm)	1.78	1.75

blocked or narrowed internal pore pathways, thereby hindering dye diffusion and leading to lower rate constants. This trade-off underscores the balance between surface functionality and pore accessibility in determining adsorption kinetics and capacity.

In conclusion, kinetic modeling reveals that 1% PEI/PAC-800 achieves high adsorption capacity through electrostatically driven, site-specific uptake. While the adsorption rate is affected by physical diffusion barriers, the enhanced surface affinity compensates for this limitation. These findings demonstrate that PEI-functionalized activated carbon is a viable candidate for the efficient removal of bulky anionic dyes from aqueous media, despite diffusion limitations.

Reusability and Desorption Behavior

To evaluate the adsorption–desorption behavior of 1% PEI/PAC-800, three consecutive adsorption–desorption cycles were conducted using RY2 as a model anionic dye. As shown in Fig. 7, the adsorbent exhibited excellent performance during the first cycle, with a maximum adsorption amount of approximately 12.8 mg and a desorption amount of 9.9 mg. The adsorption efficiency reached nearly 100%, and the desorption efficiency was as high as 83.0%, indicating strong electrostatic interactions and the presence of desorbable surface-bound species.

In the second and third cycles, the adsorption capacity declined sharply to 1.85 mg and 1.23 mg, respectively. This decrease is attributed to the partial detachment or depletion of the PEI layer during the first desorption cycle. Despite this decline, such performance degradation is not a critical limitation for powdered activated carbon (PAC)-based systems, since PAC is primarily used in batch-mode processes and is typically discarded after a single use due to its fine particle size and low recoverability [40].

What is particularly noteworthy in this study is the high desorption efficiency observed in the first cycle. Unlike conventional PAC, which often shows poor desorption behavior, 1% PEI/PAC-800 enables the effective release of adsorbed species, thereby facilitating post-adsorption treatment. This feature is especially advantageous when targeting hazardous pollutants or valuable species such as precious metal ions, which may be recovered from the spent adsorbent through desorption prior to disposal. Therefore, although 1% PEI/PAC-800 may not be suitable for repeated reuse, its high single-use performance and desorption capability offer practical benefits for selective separation and recovery applications in environmental and resource management.

Conclusion

This study evaluated the adsorption performance of PEI-impregnated powdered activated carbon (PAC) for removing the anionic dye RY2 from aqueous solutions. The 1% PEI/PAC-800 sample achieved a maximum adsorption capacity of 477.02 mg/g, about 7.4 times higher than that of raw PAC. Despite a reduction in surface area after PEI modification, the introduced amine groups enhanced electrostatic interactions under acidic conditions, improving adsorption efficiency. Kinetic analysis showed that the process followed a pseudo-second-order model, indicating

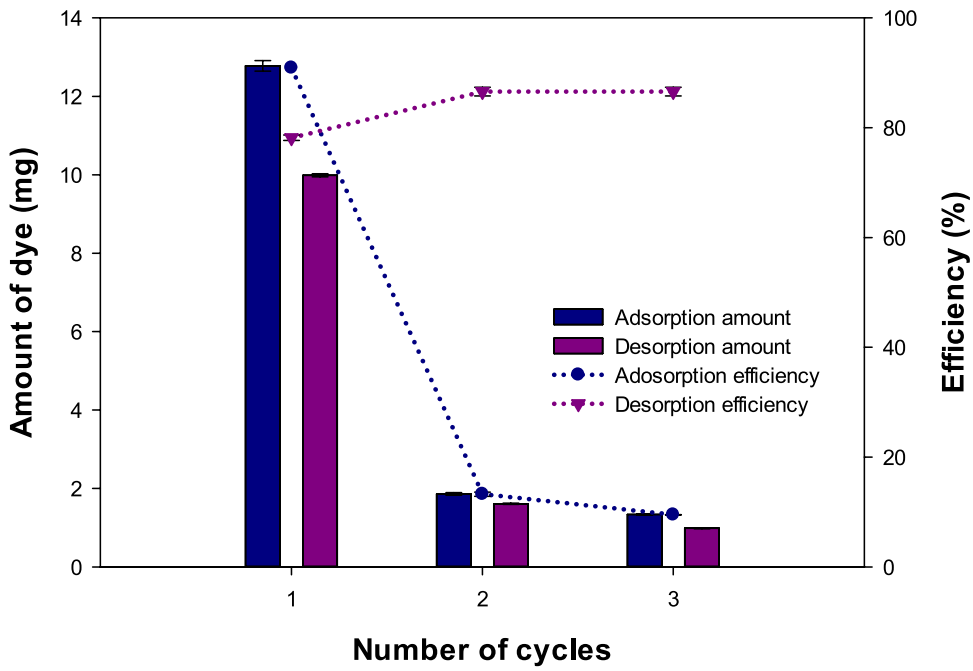


Fig. 7 Reusability of 1% PEI/PAC-800 over three adsorption–desorption cycles: Comparison of adsorption/desorption amounts and corresponding efficiencies

site-specific electrostatic interactions between the dye and protonated amine groups. Although adsorption capacity was significantly enhanced, the low rate constant suggests limited initial uptake due to diffusion constraints. Compared to microporous PAC like SPS-100, the PEI-modified adsorbent offered better dye accessibility and uptake. While reusability was limited after the first cycle, the high initial desorption efficiency and simple preparation process make the material suitable for batch-mode dye removal. In addition, the ability to release adsorbed species after use suggests potential for downstream treatment or selective separation before disposal.

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Author Contributions Su Bin Kang: Conceptualization, investigation, formal analysis, methodology, data curation, writing—original draft; Byeong-Chan Min: Formal analysis, methodology, data curation, writing—review and editing; Gwang-Il Ko: Formal analysis, methodology; Li Yong: Formal analysis, methodology; Yeoung-Sang Yun: Conceptualization, funding acquisition, supervision, writing—review and editing; Sung Wook Won: Conceptualization, funding acquisition, supervision, writing—review and editing.

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Data Availability Not applicable.

Declarations

Conflict of Interest All the authors declared that there are no conflicts of interest.

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